RACHANA KUMAR

623-286-5405 | rkuma168@asu.edu | rachanakumar-portfolio.vercel.app | linkedin.com/in/rachana-k342 | github.com/Rachana-Kumar

SUMMARY

Computer Engineering graduate student with 2+ years of experience building full-stack, AI/ML, and embedded systems, including computer vision pipelines, distributed backend services, and cloud-deployed applications. Specializes in Python, Java, C++, and REST API development. Seeking a full-time Software Engineer, Full Stack Developer, or AI/ML Engineer position.

EDUCATION

Arizona State University ***August 2024 – May 2026***
Master of Science in Computer Engineering GPA - 3.93
Relevant Courses : Artificial Intelligence, Embedded Machine Learning, Cloud Computing, Data Mining, Data Processing at Scale.

Reva University ***August 2019 – July 2023***
Bachelor of Technology in Electronics and Computer Engineering GPA - 3.72
Relevant Courses : Data Structures and Algorithms, DBMS, Computer Architecture, Digital Electronics, Operating Systems.

TECHNICAL SKILLS

Languages & Runtimes: Python, Java, C/C++, JavaScript, TypeScript, SQL, HTML, CSS, Node.js, Express.js.
Cloud & Backend: AWS, REST APIs, Apache Kafka, Redis, Docker, Node-RED, WebSocket, MQTT, MySQL, PostgreSQL.
Frameworks & Libraries: React, React Native, Expo, Angular, Spring Boot, TensorFlow, PyTorch, OpenCV, MediaPipe, Streamlit.
Tools & Platforms: Git, GitHub, Linux, JIRA, Postman, Jenkins, Supabase, CI/CD, Agile/Scrum, LLM APIs.

PROFESSIONAL WORK EXPERIENCE

Integrated Travel - Full Stack Engineer ***June 2026 – present***
- Built a full-stack maintenance management platform with 10+ REST APIs and 8+ responsive screens, enforcing secure role-based access and permissions for Residents, Maintenance Staff, and Property Managers across workflows.
- Designed a 6+ table PostgreSQL schema with trigger-based audit logging, supporting request creation, assignment, and status tracking across 4 request states with complete historical records for accountability.
- Owned authentication and request-management features from PostgreSQL schema to React UI using Git/GitHub.

Deepcept AI - AI Software Engineer ***September 2023 – May 2024***
- Developed Computer Vision pipelines for real-time intruder and object recognition using Python, OpenCV, and REST APIs.
- Integrated a remote access module into existing surveillance infrastructure via WebSocket and MQTT protocols, enabling real-time remote monitoring, reducing operator response time, and improving overall system accessibility.
- Built an end-to-end ML pipeline for Table Tennis stroke classification with TensorFlow/PyTorch, achieving real-time inference at 30fps and 92% recognition accuracy across all 8 defined stroke types on standard embedded hardware.

Telestream - Software Engineer ***February 2023 – July 2023***
- Contributed to backend development of PRISM, a real-time IP/SDI monitoring and analysis platform, processing high-throughput broadcast media streams for precise signal verification and quality compliance across all live channels.
- Delivered 20+ feature enhancements and resolved 90+ bugs across audio and waveform applications, cutting the open defect backlog by 40% and substantially improving overall product stability and long-term release confidence.
- Collaborated in Agile teams using Git, JIRA, and Jenkins for development, testing, bug tracking, and CI/CD workflows.

Bosch Ltd - QA/Test Engineer - Smart Solution Systems ***July 2022 – January 2023***
- Led QA across 5 of 14 smart campus subsystems, validating IoT integrations and automated control systems.
- Executed 300+ UAT test cases, documented defects, and collaborated with engineering teams to verify fixes.
- Identified and escalated critical issues contributing to an 80% reduction in system downtime post-deployment.

PROJECTS

Smart Menu - AI Food Ordering App - React Native, Expo, Node.js, Express.js, Groq API (Llama 3) ***Summer 2026***
- Built a solo hackathon app where an LLM turns natural-language orders into structured JSON that updates the cart, adds dietary-aware recommendations, and cuts ordering from roughly 8-10 interactions to 2-3.

Mini-Uber Backend Architecture - Java, Spring Boot, Kafka, Redis, PostgreSQL ***Spring 2026***
- Designed a distributed system for ride-dispatch using Java/Spring Boot, Kafka message queues, and Redis geo-indexing; achieved sub-100ms driver matching latency under simulated load of 10k concurrent requests.

Gym Exercise Tracker - Python, Mediapipe, Streamlit ***Fall 2025***
- Built a real-time exercise tracking app using Python, MediaPipe, and Streamlit, supporting 20+ exercise variations with automated rep-counting and form-alert notifications.

PUBLICATIONS

- "Camouflage Surveillance System," 2023 IEEE Fifth International Conference on Advances in Electronics, Computers, and Communications (ICAECC), Bengaluru, India, 2023. DOI: 10.1109/ICAECC59324.2023.10560310.
- "Design and Development of Automated Loading Dock for Local Anesthesia Deployment," Patent application filed, Intellectual Property India, Application No. 202141047093, Nov 2022.